STAFF REVIEW · H446 204 source models

Separate from the eight-page pupil workbook. This copy is in a public repository and is NOT access controlled. Check technical correctness before use.

Printed question mapping: Booklet Q1 = original online Q1; Booklet Q2 = original online Q2; Booklet Q3 = original online Q3

Do Now — source answers

1. Two variables change within the group, so only constant variables remain.

2. ¬A ∨ ¬B.

3. 10_2.

4. 1. XOR gives 1, and 1 AND 1 is 1.

I Do — source worked steps

1. Predict and explain — XOR(0,1)=1, giving Sum=1.

2. Check and extend — AND(0,1)=0, giving Carry=0.

We Do — guided CHECK

1. Predict and explain
The total is decimal 2, binary 10.
Check and extend
Sum is the low bit 0; CarryOut is the high bit 1.

You Do — source mark schemes

1. Sum=1; Carry=0.

2. $1+1+0=2=10_2$, so Sum=0, CarryOut=1.

3. Q remains 0 because it samples the current D=0 at the rising edge. Changes between edges do not update Q.

Exit Ticket — source answer

1. False. An edge-triggered D flip-flop updates Q only at its triggering clock edge.